The Derivative as a Limit and as a Rate
Answer Key
AP Calculus AB/BC

Quick Answers

1. 7 m/s
2. 6
3. 9 m/s
4. $-\frac{1}{4}$
5. (a) the rate at 5 minutes = -20, —
6. $\frac{1}{6}$
7. (a) the derivative function = $3x^2 - 2$, (b) the slope at x = 2 = 10
8. (a) the velocity function = $40 - 10t$, (b) the velocity at 2 seconds = 20, —

What is verified

10 of 12 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 5, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC
FUN-3.A – *problems 1, 2, 3, 4, 5, 6*
FUN-3.B – *problems 7, 8*
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

1. *If they got 6: divided the position at 3 seconds by 3 seconds instead of using the change in position over the change in time*
3. *If they got 7: reported the average velocity over the interval from 1 to 3 seconds instead of the instantaneous velocity at 3 seconds*

1. Cart with $s(t) = t^2 + 3t$ metres. Average velocity from $t = 1$ to $t = 3$ seconds.

Solution: An average rate is the change in output over the change in input — the slope of the line joining the two table points.

$$\frac{s(3) - s(1)}{3 - 1} = \frac{18 - 4}{2} = \frac{14}{2}$$

so the average velocity is $\boxed{7 \text{ m/s}}$

Dividing $s(3)$ by 3 instead would give 6; that is a rate measured from $t = 0$, not from $t = 1$, and it answers a different question.

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2. $f(x) = x^2$. Find $f'(3)$ by the limit definition.
Solution:

$$\begin{aligned} f'(3) &= \lim_{h \to 0} \frac{(3+h)^2 - 3^2}{h} = \lim_{h \to 0} \frac{9 + 6h + h^2 - 9}{h} \\ &= \lim_{h \to 0} \frac{h(6+h)}{h} = \lim_{h \to 0}(6+h) \end{aligned}$$

The h in the denominator cancels — that is the whole purpose of expanding first — and letting $h \to 0$ leaves $\boxed{6}$

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3. Same cart, $s(t) = t^2 + 3t$. Instantaneous velocity at $t = 3$ seconds.
Solution: $s(3) = 9 + 9 = 18$, so

$$\begin{aligned} \lim_{h \to 0} \frac{s(3+h) - 18}{h} &= \lim_{h \to 0} \frac{(3+h)^2 + 3(3+h) - 18}{h} \\ &= \lim_{h \to 0} \frac{9 + 6h + h^2 + 9 + 3h - 18}{h} = \lim_{h \to 0} \frac{h(9+h)}{h} = \lim_{h \to 0}(9+h) \end{aligned}$$

so the instantaneous velocity is $\boxed{9 \text{ m/s}}$

The contrast with problem 1 is the lesson: 7 m/s is the slope of the line joining $(1, 4)$ to $(3, 18)$; 9 m/s is the slope of the curve at the single instant $t = 3$. A student who answers 7 here has computed an interval rate where an instant was asked for.

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4. $f(x) = \dfrac{1}{x}$. Find $f'(2)$ by the limit definition.
Solution: Combine the numerator over a common denominator first:

$$\frac{\frac{1}{2+h} - \frac{1}{2}}{h} = \frac{\frac{2-(2+h)}{2(2+h)}}{h} = \frac{-h}{2(2+h)} \cdot \frac{1}{h} = \frac{-1}{2(2+h)} \qquad (h \neq 0).$$

Letting $h \to 0$ gives $\dfrac{-1}{2(2)} = -\dfrac{1}{4}$. $\boxed{-\tfrac{1}{4}}$

The negative sign is the graph telling the truth: $y = 1/x$ falls as x increases, so its slope at $x = 2$ must be negative.

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5. Draining tank, $V(t) = 60 - 2t^2$ litres. (a) $V'(5)$ by the limit definition. (b) What does it mean?

Solution (a): $V(5) = 60 - 2(25) = 10$, so

$$V'(5) = \lim_{h\to 0} \frac{[60 - 2(5+h)^2] - 10}{h} = \lim_{h\to 0} \frac{60 - 2\left(25 + 10h + h^2\right) - 10}{h}$$
$$= \lim_{h\to 0} \frac{-20h - 2h^2}{h} = \lim_{h\to 0}(-20 - 2h)$$

which gives $\boxed{-20}$

Solution (b) — instructor-judged, no machine check. The sentence must read the sign and the size separately. The sign is negative, so the tank is *losing* water at that moment. The size is 20, and the units are litres per minute because the output is litres and the input is minutes. So: at $t = 5$ minutes the volume is falling at 20 litres per minute, at that instant. Full credit needs the direction, the number with its rate units, and the idea of an instant rather than a total. Do not accept “20 litres” with no rate unit.

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6. $f(x) = \sqrt{x}$. Find $f'(9)$ by the limit definition.

Solution: Multiply by the conjugate of the numerator:

$$\frac{\sqrt{9+h} - 3}{h} \cdot \frac{\sqrt{9+h} + 3}{\sqrt{9+h} + 3} = \frac{(9+h) - 9}{h\left(\sqrt{9+h} + 3\right)} = \frac{1}{\sqrt{9+h} + 3} \qquad (h \neq 0).$$

As $h \to 0$ this tends to $\dfrac{1}{3+3}$. $\boxed{\tfrac{1}{6}}$

A tangent slope of about 0.17 is what the graph of $\sqrt{x}$ should look like at $x = 9$ — rising, but slowly.

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7. $g(x) = x^3 - 2x$. (a) $g'(x)$. (b) The slope at $x = 2$.

Solution (a): Differentiate term by term with the power rule — $\frac{d}{dx}x^n = nx^{n-1}$, which is what the limit definition produces once and for all:

$$g'(x) = 3x^2 - 2. \qquad \boxed{3x^2 - 2}$$

Solution (b): The derivative function is a machine for slopes; feed it the input. At $x = 2$: $3(2)^2 - 2 = 12 - 2$. $\boxed{10}$

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8. Challenge. $h(t) = 40t - 5t^2$ metres. (a) $h'(t)$. (b) Velocity at $t = 2$. (c) Meaning of the velocity at $t = 5$.

Solution (a): Differentiating term by term,

$$h'(t) = 40 - 10t. \qquad \boxed{40 - 10t}$$

Solution (b): At $t = 2$: $40 - 10(2) = 40 - 20$, so the ball is rising at $\boxed{20}$ metres per second.

Solution (c) — instructor-judged, no machine check. Substituting gives $h'(5) = 40 - 50 = -10$. The sentence must say that at $t = 5$ seconds the ball is moving *downward* at 10 metres per second — the negative sign is direction, the 10 is speed, and the statement is about that instant, not about distance travelled. Accept “falling at 10 m/s at $t = 5$”. Do not accept “the ball has gone down 10 metres”, which confuses a rate with a displacement.